Activity 3: Depreciation

Depreciation is the loss of value of assets, machinery or equipment through age or use.

There are 2 ways to calculate depreciation:

- Straight line depreciation $\boldsymbol{F = P(1 - i.n)}$
- Reducing balance depreciation $F = \boldsymbol{P(1 - i)^n}$

Exercise on Depreciation:

1. Stephanie buys some computer equipment valued at R250 000. This equipment depreciates in value by 2,5%p.a. on a straight line basis, Calculate the value of the computer equipment at the end of

   1) 3 years 2) 5 years 3) 10 years

   R231250 R218750 R187500

2. Amy owns a truck that cost R555 000. Calculate the value of the truck at the end of 5 years if the depreciation is calculated at 14,5%p.a. on a reducing balance.

   $A = 555000(1 - 0{,}145)^5 = R253585$

Activity 4: Exchange rates

https://www.siyavula.com/read/maths/grade-10/finance-and-growth/09-finance-and-growth-04

Read through the information on the Siyavula site (9.5 "Foreign Exchange Rates") and worked example 12.

Scroll down to **exercise 9.6**. (do questions 1, 4,6,8). Mark and correct your work.

Exercise on Exchange rates:

The current exchange rates during Covid-19 as at 20 May 2020 are as follows:

. R22,30 to the British £, R19,93 to the Euro and R18,23 to the US $

a) How many pounds sterling can be bought for R500? R22,42
b) How many rands would be paid for 350 Euros? R6975,50
c) How many dollars can be bought for R600? R32,91
d) How many dollars must be paid to purchase 200 Euros? (Hint: convert to rands first)

€ 200 = R 3986 = $ 218,65

3. **Extra exercise on Population growth:**

a. A town has a population of 65320. The average growth rate of the population is 8,2%p.a.

Determine the expected population of the town in 20 years' time assuming the growth rate is constant.

Population $= 65320(1+0{,}082)^{20} = 315930{,}3812$

b. The number of foreign tourists in the Cape Winelands in 2011 was 2 145 637.
That number is expected to increase annually by 4%p.a. Calculate how many tourists are expected in the Winelands in 2020. (Covid-19 has put paid to that I am sure!)

Tourists $= 2145637(1+0{,}04)^{9} = 3053910{,}487$

c. In 1980 the number of Zebra in the Kruger National Park was estimated to be 22 500.
In 2010 the number was estimated to be 30 000. Calculate the average percentage increase per annum in the Zebra population over the 30-year period.

$30000 = 22500(1+i)^{30}$

$4/3 = (1+i)^{30}$

$i = 0{,}96\%$ increase

d. A game farm has a herd of 64 impala. The manager calculated that the farm can sustain a herd of 250 impala and thereafter he will sell surplus stock. If the herd grows at 17%p.a., calculate how long he will have to wait before selling off stock. (Use the trial and error technique from your work with exponents).

$250 = 64(1+0{,}17)^{n}$

$\frac{125}{32} = (1{,}17)^{n}$

$3{,}90625 = 1{,}17^{n}$

$n = 8{,}67$ & approx

e. A dairy farmer has a herd of 540 dairy cows. He plans to increase his herd to 1200 using a breeding programme. The herd is expected to increase by 9%p.a. on the programme. Calculate how long it will take to achieve the target.

$1200 = 540(1+0{,}09)^{n}$

$\frac{20}{9} = (1{,}09)^{n}$

$n = 9{,}266$ approx

or 9yr 3months

Mixed Revision Exercise MAS Pg 239

Activity 7: "Adjusted" Compound Interest Rates

So far when working with compound interest rates, the rates have been given **per annum (i.e. per year).**

E.g. R1000 is invested in a compound growth account for 5 years at 6% p.a. How much money will be in the account at the end of the 5 years?

$A = 1000(1+0{,}06)^{5}$

$= R1338{,}23$

However, compound interest can be awarded **daily, weekly, monthly, quarterly** etc. and we have to **adjust our formula** to cover these options.

$A = P(1+i)^{n}$ n becomes the number of times interest is given

i becomes the adjusted interest per time

Complete the following:

1. 8% p.a. compounded quarterly

Each quarter the interest rate will be : $\frac{0{,}08}{4} = 0{,}02$

Interest will be awarded 4 times in a year.

R 4000 is invested in an account for 2 years, interest is 8%p.a. compounded quarterly. How much is in the account at the end of the 2 years?

$P = 4000$ $i = \frac{0{,}08}{4}$ $n = 4 \times 2$

$$A = 4000\left(1 + \frac{0{,}08}{4}\right)^{4\times 2}$$

$$= R4686{,}64$$

2. 14%p.a. compounded monthly

Each month the interest rate will be : $\frac{0{,}14}{12}$

Interest will be awarded 12 times in a year.

R3500 is invested in an account for 10 years, interest is 14% compounded monthly. How much money is in the account at the end of the 10 years?

$P = 3500$ $i = \frac{0{,}14}{12}$ $n = 10 \times 12$

$$A = 3500\left(1 + \frac{0{,}14}{12}\right)^{10\times 12}$$

$$= R14078{,}65$$

3. 7% p.a. compounded daily

Each day the interest rate will be : $\frac{0,07}{365}$

Interest will be awarded 365 times a year.

R 2500 is invested for 3 years in an account where interest is 7% p.a. compounded daily. How much is in the account after 3 years?

$P = 2500$ $i = \frac{0,07}{365}$ $n = 365 \times 3$

$A = 2500\left(1 + \frac{0,07}{365}\right)^{365 \times 3}$

$= R3084,13$

4. 23 % p.a . compounded semi-annually

Semi-annually the interest rate will be : $\frac{0,23}{2}$

Interest will be awarded 2 times a year.

R 4000 is invested in an account for 18 months. The interest is 23%p.a. compounded semi-annually. How much money is in the account at the end of the 18 month period?

$P = 4000$ $i = \frac{0,23}{2}$ $n = 2 \times 1,5$

$A = 4000\left(1 + \frac{0,23}{2}\right)^{2 \times 1,5}$

$= R5544,78$

5. Going back to the first problem…. R1000 grows to R1338,23 if invested for 5yrs at 6%p.a.

How much would it grow to if invested for 5 years at 6%p.a. compounded monthly?

$A = 1000\left(1 + \frac{0,06}{12}\right)^{12\times 5}$

$= R1348,85$

How much would it grow to if invested for 5 years at 6% p.a. compounded daily?

$A = 1000\left(1 + \frac{0,06}{365}\right)^{365\times 5}$

$= R1349,83$

Which of these investment options would you choose?

Compounded daily

SUMMARY

Compounded:	**Adjustment for n**	**Adjustment for *i***
Yearly	n	i
Semi- annually Half-yearly	$n \times 2$	$\frac{i}{2}$
Quarterly	$n \times 4$	$\frac{i}{4}$
Month	$n \times 12$	$\frac{i}{12}$

<u>Exercise on "Adjusted" Compound Interest Rates</u>:

1. R24 000 is deposited in a savings account. The interest paid is 14,5% p.a. compounded monthly. Calculate the value of the investment after:

   a) 5 years b) 10 years c) 15 years

   d) What do you notice about the relative increase in the investment every 5 years?

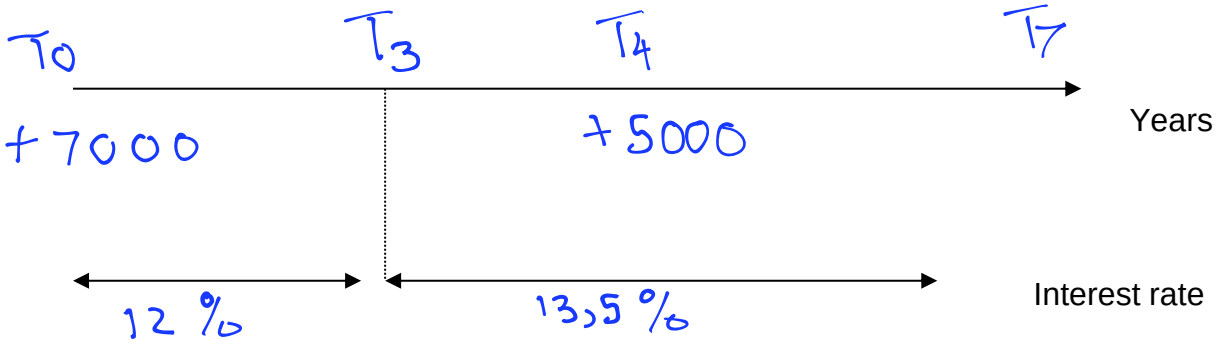
Activity 8: Time Lines

A time line helps to summarize the information and provides a visual representation.

HINT: Try and work each option to the end of the time period given, combine at the end.

EXAMPLE 1

R7000 is deposited in a savings account, four years later another R5000 is added to the savings. Calculate the value of the savings at the end of 7 years if the interest rate is 12% p.a. for the first 3 years, and then is increased to 13,5%p.a. for the remaining period.



$$A = 7000(1+0,12)^3(1+0,135)^4 + 5000(1+0,135)^3$$

$$= R23631,26$$

Exercise on Time Lines

1) R3 000 is deposited in a savings account for 5 years. The interest for the first two years is 9,5% p.a. compounded quarterly, and it is then increased to 10,2%p.a. compounded monthly. Calculate the amount of money accumulated at the end of 5 years.

2) R1 400 is deposited into a savings account, and after 9 months, another R2 100 is added to the savings. If the interest rate was quoted as 8,25% p.a. compounded quarterly, calculate the total amount in the account at the end of two years.

3) R28 000 is invested at 15,8%p.a. compounded monthly for the first three years, and then at 14%p.a. compounded daily for the next three years, Calculate the value of the investment at the end of 6 years.

ANSWERS 1) R4 909,14 2) R3974,06 3) R68 239,98

Consolidation Exercise:

1. R6 578 is invested for 48 months at 9,5%p.a. simple interest . How much is in the account at the end of the 48 months? R 9077,64

2. Mrs Smith buys a fridge and deep freezer for R8 700. She pays for them over 3 years on a hire purchase agreement. The interest rate is 12%p.a. based on the full purchase price over three years.

   a) How much will she end up paying for these items over the 3 years? R11 832

   b) How much will she pay each month if an insurance fee of R22 is added to each month's payment? R 350,67

3. An investment triples in value after 5 years if compounded annually. At what rate was the investment compounded? $i = 24,57\%$ p.a.

4. I deposit R6000 in the bank for my 7 year old daughter's 18th birthday. I give her an amount of R20 000 on her birthday. What rate did I receive on my investment, if simple interest was calculated? $i = 21,21\%$ p.a

5. R 5 650 is invested in account for 5 years. Interest is set at 10,5% compounded weekly. How much money will be in the account at the end of the 5 year period? R 9546,04

6. An investment of R12 500 earned 12% p.a. for 4 years compounded annually. The interest rate then dropped to 11% p.a. for the next 3 years. Calculate the final amount in the account after the 7 years. R 26 899,92

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